

Rでの解析の流れ

```
> 18-12A_jyuku <- read.table('./gakusyuu_jyuku.csv', header=T, sep=',')
> head(18-12A_jyuku)
  ID kokugo syakai rika eigo suugaku
1  A     93    100   89   84      77
2  B    100     98   89   95      86
3  C     84     84   99   85     100
4  D     70     73   92   66      77
5  E     70     72   89   66      75
6  F     66     68   95   57      82

> 18-12A_jyuku_fact <- factanal(18-12A_jyuku[,2:6], factors=2, scores='regression', rotation='varimax')
> 18-12A_jyuku_fact

Call:
factanal(x = df[, data_cols], factors = factors, scores = scores,      rotation = rotation)

Uniquenesses:
  kokugo  syakai    rika    eigo suugaku
    0.005   0.109   0.118   0.290   0.005

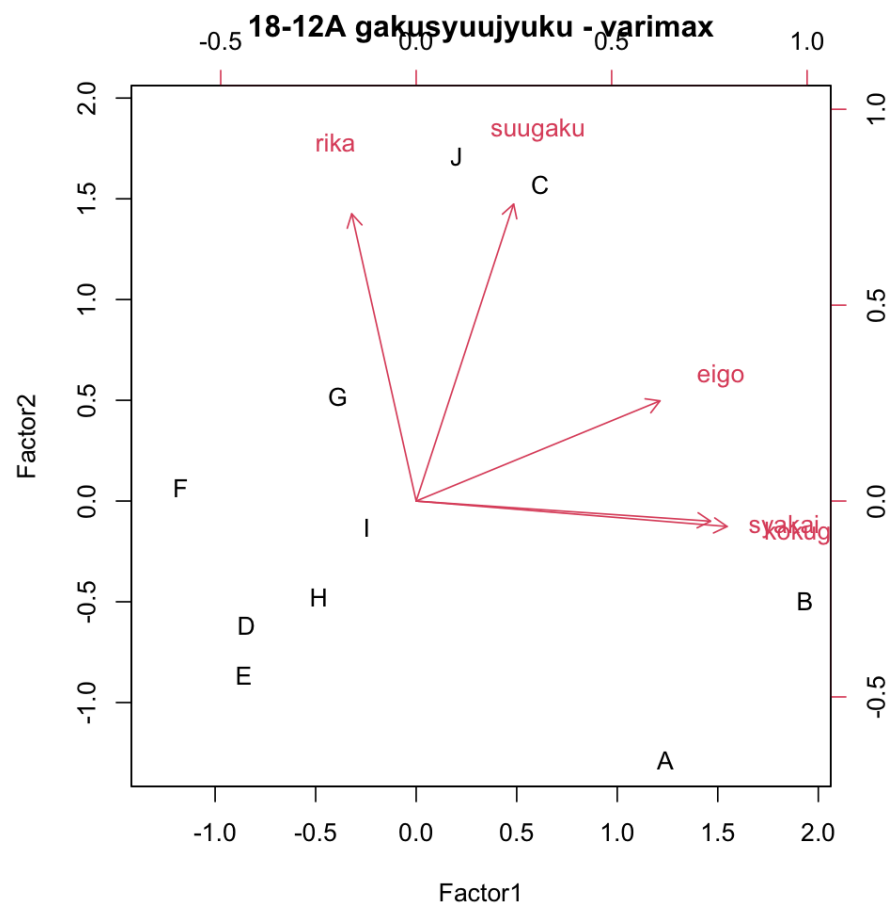
Loadings:
      Factor1 Factor2
kokugo   0.994
syakai   0.942
rika    -0.206   0.916
eigo     0.780   0.320
suugaku   0.312   0.948

      Factor1 Factor2
SS loadings    2.622   1.851
Proportion Var  0.524   0.370
Cumulative Var  0.524   0.895

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 1.34 on 1 degree of freedom.
The p-value is 0.247

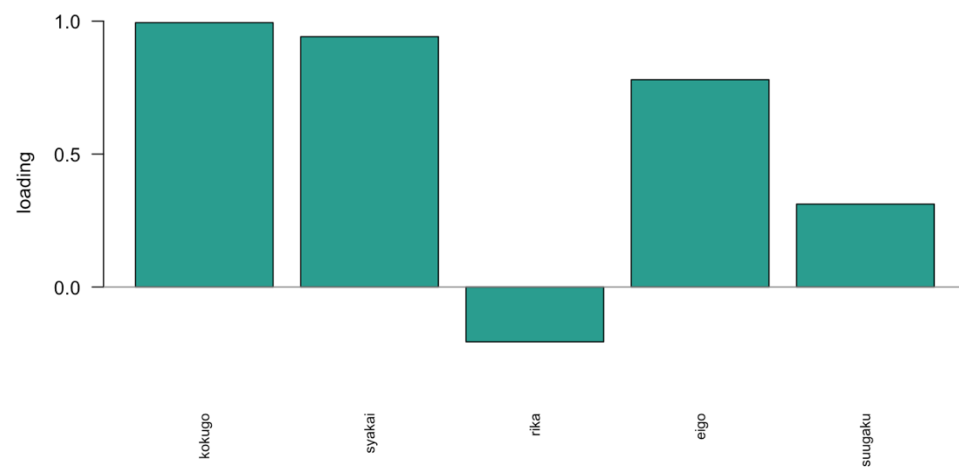
> 18-12A_jyuku_fact$loadings[,1:2]
      Factor1    Factor2
kokugo  0.9942328 -0.08128666
syakai  0.9415011 -0.06434947
rika   -0.2060710  0.91640593
eigo    0.7796112  0.31970998
```

因子得点と因子負荷量

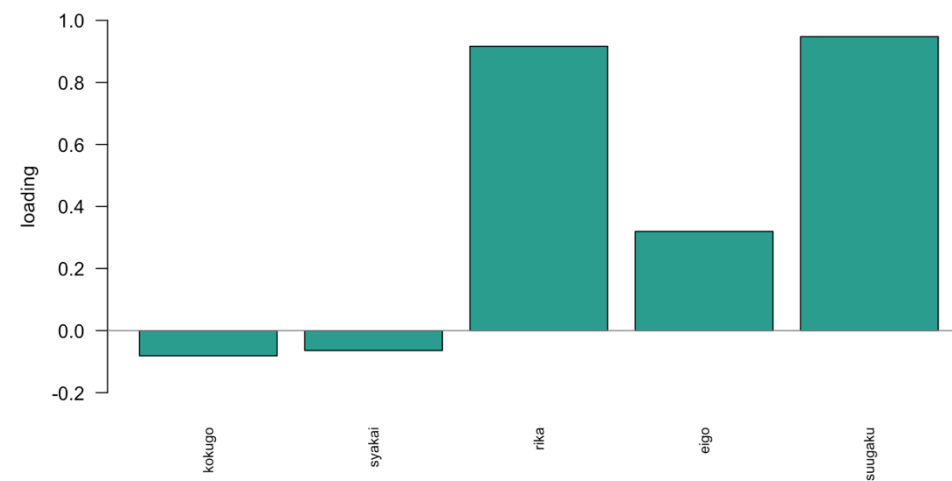


因子負荷量の棒グラフ

18-12A gakusyuuujyuku - varimax Factor1 loadings



18-12A gakusyuuujyuku - varimax Factor2 loadings



ソースコード (1/2)

Class13_rp18-12A_02.R

```
1 setwd("/Users/kakigoori/Desktop/data/13")
2
3 dir.create("./answers/plots_pdf", recursive=TRUE, showWarnings=FALSE)
4
5 jyuku <- read.table("./gakusyuu_jyuku.csv", header=T, sep=",")
6
7 jyuku
8
9 head(jyuku)
10
11 jyuku_fact <- factanal(jyuku[,-1], factors=2, scores="regression", rotation="varimax")
12
13 str(jyuku_fact)
14
15 jyuku_fact
16
17 jyuku_fact$loadings[,1:2]
18
19 apply(jyuku_fact$loadings, 1, function(x){1-sum(x^2)})
20
21 jyuku_fact$scores
22
23 pdf("./answers/plots_pdf/18-12A_jyuku_biplot.pdf", width=8.5, height=6.2)
24 biplot(jyuku_fact$scores, jyuku_fact$loadings, xlab=jyuku[,1])
```

ソースコード (2/2)

Class13_rp18-12A_02.R

```
25 dev.off()
26
27 pdf("./answers/plots_pdf/18-12A_jyuku_factor1_bar.pdf", width=8, height=5.2)
28 barplot(jyuku_fact$loadings[,1], cex.names=0.8, las=2, ylim=c(-0.3,1.1))
29 abline(h=0, col="gray50")
30 dev.off()
31
32 pdf("./answers/plots_pdf/18-12A_jyuku_factor2_bar.pdf", width=8, height=5.2)
33 barplot(jyuku_fact$loadings[,2], cex.names=0.8, las=2, ylim=c(-0.3,1.1))
34 abline(h=0, col="gray50")
35 dev.off()
```